

Artificial Intelligence – AI Agent Assignment

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Semester: 6th **Course:** Artificial Intelligence

1. AI Agent

Agent Name: TravelMate – Smart Travel Planning Agent

TravelMate is a simple AI agent that helps a user choose a suitable travel destination. It takes information such as budget, number of days, and travel interest, and then recommends a destination based on the available choices.

2. Graphical Representation

User Budget, days, interest	Sensors / Input Collect user requirements	TravelMate Agent Analyze and select destination	Actuators / Output Recommendation, cost, activities
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3. Agent Formulation

Component	Description
Goal	Recommend a suitable travel destination for the user.
Input	Budget, number of days, and travel interest.
Process	Compare user requirements with destination information.
Output	Best matching destination with estimated cost and activities.

4. Environment Types

- **Partially observable:** The agent does not know all the user's preferences.
- **Stochastic:** Travel conditions, prices, and weather can change.
- **Sequential:** The current recommendation can affect later travel decisions.
- **Dynamic:** Travel information may change with time.
- **Mostly discrete:** Destinations and activities are selected from choices.
- **Mostly single-agent:** The main decision maker is the travel agent, although other people and services can affect travel.
- **Partly unknown:** Some information such as future prices or weather may not be known.

5. PEAS Framework

PEAS	TravelMate
Performance Measure	Good destination match, suitable budget, suitable number of days, and useful recommendation.
Environment	Destinations, transport, hotels, activities, prices, weather/season, and user requirements.
Actuators	Display recommended destination, estimated cost, activities, and explanation.
Sensors	User's budget, number of days, interests, and available travel information.

6. Conclusion

TravelMate is a simple AI travel recommendation agent. It collects user requirements, checks the available travel options, and gives a suitable recommendation. This shows how an AI agent can sense information, make a decision, and produce an action or result.